

MEDICAL APPOINTMENT SCHEDULER A C++ PROGRAMMING MINI PROJECT

Features Required

- Add Patient Details (Patient ID, Name, Age, Disease)
- Book Appointments
- Cancel Appointments
- Search Patient Details by Patient ID or Name
- Display Appointment Queue
- Store records permanently using file handling

Concepts to be Used

- Classes and Objects
- Queues
- Arrays
- Loops
- Conditional Statements
- Functions
- File Handling

Technology Used

- C++ Programming Language

Output

- Console-based formatted MEDICAL APPOINTMENT SCHEDULER
- Permanent data storage using files (patients.txt, appointments.txt)

1. INTRODUCTION

- The Medical Appointment Scheduler is a simple C++ language-based mini project designed to manage hospital or clinic appointments efficiently.
- It helps in storing patient information, booking appointments, managing waiting queues, and maintaining records using file handling concepts.

- This project is developed for academic learning purposes and demonstrates how object-oriented programming concepts can be applied to real-world healthcare systems.

2. OBJECTIVES

- To maintain patient records digitally
- To automate appointment scheduling
- To reduce manual work and waiting time
- To provide quick patient search facilities
- To demonstrate queue implementation using C++

3. SYSTEM FEATURES

- 3.1 Add Patient Details – Stores patient details such as ID, Name, Age, and Disease.
- 3.2 Book Appointments – Patients are added to the appointment queue.
- 3.3 Cancel Appointments – Existing appointments can be cancelled.
- 3.4 Search Patient Records – Search by Patient ID or Name.
- 3.5 Display Queue – Displays all scheduled patients in order.

4. TECHNOLOGY & CONCEPTS USED

- Classes and Objects
- Queues for appointment management
- Loops for menu handling
- Conditional Statements
- Functions for modular programming
- File Handling for permanent storage

5. DATA STRUCTURE

- Example Class Structure:
- `class Patient { int id; string name; int age; string disease; };`

6. WORKFLOW

- Enter patient details
- Store data in files
- Book appointments

- Display appointment queue
- Search or cancel appointments

7. FILE HANDLING APPROACH

- Records are stored using file handling
- patients.txt – Stores patient details
- appointments.txt – Stores appointment records
- Data remains stored even after program termination

8. SAMPLE OUTPUT (CONSOLE)

- =====
- MEDICAL APPOINTMENT SCHEDULER
- =====
- Patient ID : 101
- Name : Rahul
- Age : 22
- Disease : Fever
- Appointment Booked Successfully

9. ADVANTAGES

- Reduces patient waiting time
- Easy appointment management
- Improves accuracy
- Quick search facility
- Permanent record storage

10. LIMITATIONS

- Console Based Interface
- Limited Security
- Single User Access

11. FUTURE ENHANCEMENTS

- Graphical User Interface (GUI)

- Database Integration
- Online Appointment Booking
- SMS and Email Notifications

12. APPLICATIONS

- Hospitals
- Clinics
- Medical Centers
- Diagnostic Laboratories
- Healthcare Training Institutes

13. CONCLUSION

- The Medical Appointment Scheduler is a simple and effective C++ based mini project.
- It demonstrates queue management, file handling, and object-oriented programming concepts.
- This project reduces manual work and is useful for beginners to understand real-world healthcare systems.

Project Information

- Project Name: MEDICAL APPOINTMENT SCHEDULER
- Technology: C++ Programming Language
- Type: Console Based Mini Project

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